Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	4
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Evaluation, Standardization and Normalization To understand and implement **evaluation techniques, standardization, and normalization** for data preprocessing and performance assessment in Machine Learning models.
Problem Definition	In Machine Learning, evaluating model performance and preparing data properly are essential steps for building accurate models. Different evaluation metrics are used depending on the type of problem, such as **classification** or **regression**. Classification problems use metrics like **Accuracy, Precision, Recall, and F1-score**, while regression problems use metrics like **Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 score**. Data preprocessing techniques such as **standardization** and **normalization** are used to scale features so that machine learning algorithms can perform better. This practical demonstrates these techniques and compares their results.
Theory (100 words)	Evaluation metrics are used to measure the performance of machine learning models. For classification problems, common metrics include accuracy, precision, recall, and F1-score, which help determine how well the model predicts class labels. For regression problems, metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 score measure prediction accuracy. Data preprocessing techniques such as normalization and standardization are used to scale numerical data. Normalization scales data between 0 and 1, while standardization transforms data to have a mean of 0 and standard deviation of 1. These techniques improve model performance and training stability.

Procedure and Execution

(100 Words)

Algorithms:

Algorithm for Evaluation Metrics

1. **Start the program.**
2. **Import required libraries.**
3. **Define actual and predicted values.**
4. **Calculate classification metrics such as accuracy, precision, recall, and F1-score.**
5. **Calculate regression metrics such as MSE, RMSE, MAE, and R^2.**
6. **Display the evaluation results.**

Algorithm for Standardization

1. **Start the program.**
2. **Import the required library for preprocessing.**
3. **Define the dataset.**
4. **Apply StandardScaler to transform the data.**
5. **Display the standardized dataset.**

Algorithm for Normalization

1. **Start the program.**
2. **Import preprocessing library.**
3. **Define the dataset.**
4. **Apply MinMaxScaler to scale the values between 0 and 1.**
5. **Display the normalized dataset.**

Code:

```
import numpy as np
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score
from sklearn.metrics import mean_squared_error,
mean_absolute_error, r2_score
from sklearn.preprocessing import StandardScaler, MinMaxScaler

# Sample classification data
y_true = [1,0,1,1,0]
y_pred = [1,0,0,1,0]

print("Classification Metrics")
print("Accuracy:", accuracy_score(y_true, y_pred))
print("Precision:", precision_score(y_true, y_pred))
print("Recall:", recall_score(y_true, y_pred))
print("F1 Score:", f1_score(y_true, y_pred))

# Sample regression data
y_true_reg = [3, -0.5, 2, 7]
y_pred_reg = [2.5, 0.0, 2, 8]

print("\nRegression Metrics")
print("MSE:", mean_squared_error(y_true_reg, y_pred_reg))
print("RMSE:", np.sqrt(mean_squared_error(y_true_reg,
y_pred_reg)))
```

```
print("MAE:", mean_absolute_error(y_true_reg, y_pred_reg))
print("R2 Score:", r2_score(y_true_reg, y_pred_reg))

# Dataset for scaling
data = np.array([[10],[20],[30],[40],[50]])

# Standardization
scaler = StandardScaler()
standardized_data = scaler.fit_transform(data)

print("\nStandardized Data")
print(standardized_data)

# Normalization
normalizer = MinMaxScaler()
normalized_data = normalizer.fit_transform(data)

print("\nNormalized Data")
print(normalized_data)
```

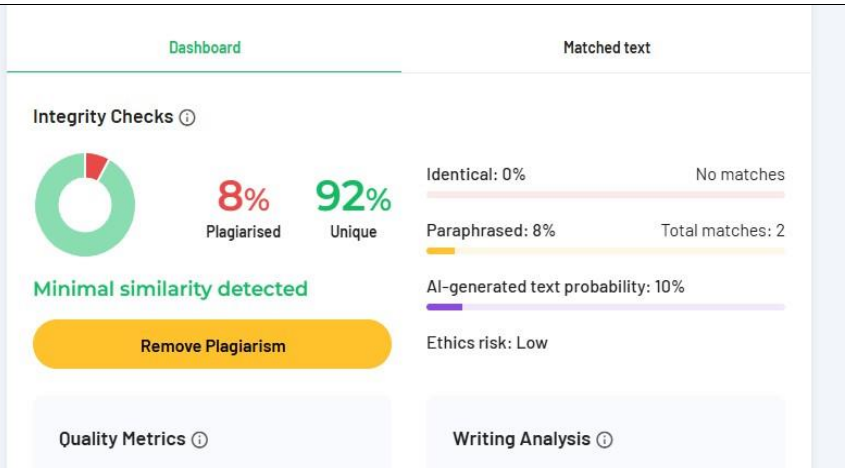
Output:

```
Classification Metrics
Accuracy: 0.8
Precision: 1.0
Recall: 0.6666666666666666
F1 Score: 0.8

Regression Metrics
MSE: 0.375
RMSE: 0.6123724356957945
MAE: 0.5
R2 Score: 0.9486081370449679

Standardized Data
[[-1.41421356]
 [-0.70710678]
 [ 0.        ]
 [ 0.70710678]
 [ 1.41421356]]

Normalized Data
[[0.  ]
 [0.25]
 [0.5 ]
 [0.75]
 [1.  ]]
```

Output Analysis

Comparison and Analysis
Normalization scales data between **0 and 1**, which is useful when features have different ranges. Standardization transforms data so that it has **mean = 0 and standard deviation = 1**, which is suitable for algorithms like **SVM and Logistic Regression**. Evaluation metrics help measure the performance of machine learning models and determine whether the model is accurate and reliable.

Output Analysis
The program calculates both **classification and regression evaluation**

	metrics to measure model performance. It also demonstrates **standardization and normalization** techniques used for data preprocessing. These transformations help machine learning algorithms perform better by ensuring that all features are on a similar scale.
Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The practical successfully demonstrated evaluation techniques and preprocessing methods in machine learning. Classification and regression metrics were used to evaluate model performance, while normalization and standardization were applied to scale the dataset. These techniques are important for improving model accuracy and efficiency.
Plag Report (Similarity index < 12%)	
Date	02/02/26

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